
JobTrace

A plain-language guide to how your tracker works

This guide explains, in everyday terms, what JobTrace does, where your information is kept, and how it checks your email automatically. No technical background is needed to read it.

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What JobTrace Is

JobTrace is a personal job-application tracker that lives entirely on your own computer. Think of it as a private notebook that keeps a tidy record of every job you've applied to — the company, the role, what stage things are at, and everything that has happened along the way.

Nothing about it depends on the internet or on any outside company. Your information is never uploaded anywhere, stored on someone else's server, or shared with a third party. It stays on this computer, in a folder you can open and look at any time.

The two parts working together

- **A dashboard you look at.** A clean, easy-to-read screen that shows your applications, how many you've heard back from, and simple charts of your progress.
- **A helper that reads your email.** A background process that checks your inbox every so often, recognizes recruitment-related emails, and quietly updates your tracker so you don't have to type everything in by hand.

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Where Your Information Lives

No cloud, no company, no subscription

Everything JobTrace knows is stored in one place: a single data file kept in a folder on this computer, alongside the program itself. There is no external database, no cloud account, and no company on the other end that could see or lose your information.

Why keeping it local matters

Because everything stays in one file on one computer, there's no risk of two different copies of your data drifting apart or overwriting each other — the kind of problem that can happen with cloud-synced or shared documents. There is exactly one, current, trustworthy copy of your data at all times.

A quick tour of what's kept where

- **Your applications.** Every company, role, current stage, and outcome you track — the heart of the system.
- **Your timeline.** A running history for each application: when you applied, when you heard back, every interview, every update.
- **A record of email checks.** A simple note of when the automatic email check last ran, and a short activity log each time it runs — so you can always see exactly what it found.
- **A "needs your attention" list.** Emails the automatic check wasn't fully confident about are set aside here rather than guessed at, so you can review them yourself.
- **Automatic backups.** Whenever the program needs to upgrade how it stores information, it saves a safety copy first, without being asked.

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Understanding Your Applications

Every application you track has two separate pieces of information, kept apart on purpose, so that a rejection never erases how far you actually got.

What it tracks	In plain terms
Stage	Where the application currently sits — Applied, Screening, Assessment, an interview round, Offer, or Closed.
Outcome	How it's going — Pending, a positive signal, a rejection, an accepted offer, or one you withdrew from yourself.

For example, an application that reached the final interview and was then rejected still shows “Final Interview” in its history — the rejection doesn't erase how far it got, it's simply recorded as the outcome.

What “heard back” means

The dashboard's “Heard Back” figure counts anything beyond a plain application confirmation — a screening call, an assessment, or an interview invitation all count as having heard back, not just an explicit yes-or-no reply. It's split into positive signals and negative ones, so you can see the balance at a glance.

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The Dashboard

Opening JobTrace — with the shortcut on your desktop — quietly starts everything in the background and opens the dashboard in your browser. Nothing technical appears on screen; you just see the tracker itself.

What you'll see

- **Summary cards.** Total Applications, Heard Back (broken into positive and negative), Interviews, and Offers — the headline numbers, always visible.
- **The full list.** Every application you're tracking, sortable and searchable, with its current stage and outcome.
- **Charts.** How many applications you've sent over time, a breakdown of where everything currently stands, and a view of your overall journey.
- **An email-status indicator.** A small pill in the top bar shows when the automatic email check last ran and a short summary of what it found.

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How the Automatic Email Check Works

Read-only, careful, and cautious by design

JobTrace can look through your email for recruitment-related messages and update your tracker on

its own. This connection is deliberately narrow: it can only look at your email, and nothing else.

It can never take action in your inbox

The email check is only ever allowed to search for and open messages. It cannot send an email, write a reply, delete or archive anything, change any label, mark anything as read or unread, or unsubscribe from anything on your behalf. Those abilities simply aren't given to it — there's no setting to accidentally enable them.

What happens during a check

- **1.** It picks up where it left off, only looking at messages received since the last check — it doesn't re-read your whole inbox every time.
- **2.** It looks for recruitment-related messages, including ones you've already moved to Trash, since it's normal to read and then tidy up an email.
- **3.** It works out what each relevant email means — an application confirmation, an interview invitation, a rejection, and so on — and matches it to the right application in your tracker, or creates a new one if it's genuinely new.
- **4.** When it isn't confident about a match, it doesn't guess. It sets the email aside in a “needs your attention” list instead of risking a mistake.
- **5.** It remembers every email it has already looked at, so re-running a check never creates duplicates or repeats an update.
- **6.** It finishes by writing a short, plain-English summary of what it found and changed — saved so you can look back at it any time.

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How Often It Runs

The email check happens automatically every 8 hours, using your computer's own built-in scheduling tool — the same feature Windows uses for routine background tasks. It only runs while you're signed in to this computer; it won't run if the computer is off or signed out.

There's nothing to maintain and nothing to pay for. It uses your existing sign-in to Claude, so there's no separate account, subscription, or ongoing cost involved.

If you ever want to check on it or turn it off

- **To see what a check found,** look at the email-status pill on the dashboard, or open the day's activity log in your JobTrace folder.
- **To pause or stop it,** open the Task Scheduler tool built into Windows (search for “Task Scheduler” in the Start menu), find the entry named “JobTrace Gmail Sync,” and disable or delete it. Doing so takes effect immediately and has no effect on your existing data.

Turning the automatic check off never affects your saved applications — it only stops new emails from being read in going forward.

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How Your Data Is Kept Safe

- **It never leaves this computer.** There's no cloud backup, no syncing service, and no outside company with a copy of your information.
- **Safety copies before any upgrade.** If the way information is stored ever needs to change, a full backup is made automatically first, without being asked.
- **No duplicate entries.** Both the dashboard and the automatic email check are built so the same action can never be recorded twice — even if a check is accidentally run twice over the same emails.
- **You can always export a copy.** The dashboard can save your entire list of applications to a plain spreadsheet-friendly file at any time, giving you an independent copy whenever you'd like one.

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Quick Reference

Opening and closing JobTrace

- **To open it,** double-click the JobTrace shortcut on your desktop. It starts quietly in the background and opens the dashboard for you.
- **To close it,** simply close the browser tab — the background helper can be left running, or stopped from the JobTrace folder if you prefer.

Checking on the automatic email sync

- The status pill at the top of the dashboard always shows the last check's time and a short summary.
- A written summary of every check is also saved in your JobTrace folder, if you'd like the fuller picture of what it looked at and found.
- Anything it wasn't sure about is listed for your review right in the dashboard — nothing is ever silently guessed.

The sections above are everything you need day to day. What follows is an optional deeper look under the hood — how the program is actually built, what each piece is called, and exactly how the automatic email check talks to Claude. Nothing here requires a technical background; every term is explained as it comes up.

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Behind the Scenes: The Building Blocks

JobTrace is built from three separate pieces that work together, plus a fourth piece — the automation — that ties into them from the outside. Each one has a plain job to do.

Piece	Its job	Built with
The website (frontend)	What you actually see and click on screen	HTML, CSS, and JavaScript — the same three building blocks every website is made of
The engine (backend)	Does the real work: saves data, runs calculations, enforces the rules	Python, a widely used programming language
The filing cabinet (database)	Stores every application and its history as one file on this computer	SQLite, a simple, file-based way of storing information
The automation	Reads Gmail on a timer and updates the filing cabinet for you	Windows' built-in scheduler + the Claude Code program

None of this runs on the internet. The “website” only exists on this computer and is never visible to anyone else, even though it looks and behaves like an ordinary website in your browser.

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The Website You See

The frontend

Everything you look at — the dashboard, the charts, the forms for adding an application — is a small website made of three kinds of files, all stored in JobTrace's frontend folder:

- **HTML** (HyperText Markup Language) lays out what's on the page — this box here, that button there. One file, `index.html`, defines the entire page.
- **CSS** (Cascading Style Sheets) decides how it all looks — the dark background, the gold highlights, the layout of the cards and charts. This lives in a single `styles.css` file.
- **JavaScript** makes the page interactive — it's what redraws the charts, opens and closes forms, and talks to the engine behind the scenes without ever needing to reload the page. This is split into several small files by purpose (one for the dashboard, one for charts, one for the Gmail status pill, and so on).

Your browser (Chrome, Edge, whichever you use) is what actually reads these files and turns them into the screen you interact with — the same way it would for any website, except these files are sitting on your own hard drive instead of being downloaded from the internet.

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The Program Doing the Work

The backend

Behind the website sits a small Python program. Python is simply the language its instructions are written in — nothing about it is visible to you day to day. When you start JobTrace, this program quietly begins running in the background and does two jobs at once:

- **1.** It hands your browser the HTML, CSS, and JavaScript files described above, so the dashboard can appear in the first place.
- **2.** It listens for requests from that page — “give me the list of applications,” “save this new one,” “what are today’s statistics” — and answers each one. This back-and-forth only ever happens between your browser and this program, both on the same computer; nothing is sent out to the internet.

Each of those requests goes to a fixed, specific address, such as `/api/applications` or `/api/stats/summary` — this fixed list of addresses is usually called an API. Think of it as a very short, very literal menu: the webpage can only ask for exactly the things on that menu, nothing else, which keeps the whole system predictable and easy to reason about.

The program is split into a few files, each with one job

File	What it's responsible for
<code>server.py</code>	Listens for requests from the browser and routes each one to the right place
<code>repository.py</code>	The actual rules — how to create, update, and search applications; how the dashboard numbers are calculated
<code>database.py</code>	Opens and manages the data file itself; sets it up the first time it's used
<code>gmail_sync.py</code>	Keeps track of which emails have already been checked, and when the last email check ran
<code>constants.py</code>	The fixed lists everything else relies on — the valid stages, outcomes, and categories

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Where Your Data Actually Lives

The database

Every application, every stage change, every event on a timeline is stored in one single file: `data/applications.db`. It uses a format called SQLite — essentially a very organized, structured filing

cabinet built into a single file, rather than a spreadsheet or a folder of loose documents. Nothing needs to be installed or run separately for this to work; Python can read and write it directly.

Inside that one file, your information is split into two related tables: one row per application (company, role, current stage, outcome), and one row per event on that application's timeline (an interview invitation, a rejection, and so on) — linked back to the application it belongs to.

Why the dashboard and the email check never collide

The data file runs in what's called “WAL mode” (write-ahead logging) — a setting that lets one process read the file while another is writing to it, safely, at the same moment. That's what allows you to have the dashboard open in your browser at the exact same time the scheduled email check is quietly updating it in the background, with no risk of the two interfering with each other.

The other files alongside it

- **gmail_sync_state.json**, a small file recording the date and time of the last successful email check.
- **unresolved_gmail_items.json**, the “needs your attention” list of emails the check wasn't confident enough to act on by itself.
- **sync_logs**, a folder holding one plain-text report per email check, so you can look back at exactly what any run found.
- **The backups folder**, which automatically receives a full safety copy of applications.db any time the structure of the data needs to change.

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From Click to Saved

What actually happens behind the scenes

As a concrete example, here's exactly what happens when you add a new application by hand on the dashboard:

1. You fill in the form in your browser and click Save. The page's JavaScript gathers what you typed.
2. It sends that information, on this computer only, to the Python program's `/api/applications` address — a request that essentially says “please create this application.”
3. The Python program checks that everything makes sense (a valid stage, a valid outcome, no missing required details), then writes a new row into `applications.db`.
4. It sends a confirmation back to the page, which then redraws itself to show the new application — all without the page ever reloading.

Every action in JobTrace — editing a stage, adding a timeline event, deleting an application, exporting your data — follows this same basic pattern: the page asks, the Python program checks and carries it out, the data file is updated, and the page reflects the result.

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How Claude Talks to the Scheduled Task

The automation, step by step

This is the piece that runs without you doing anything, so it's worth spelling out exactly how it works —

there's no mystery to it, just a chain of ordinary steps.

- **1.** Every 8 hours, Windows' own Task Scheduler — a feature built into every Windows computer for running routine background tasks — starts a small script called `run_gmail_sync.bat`.
- **2.** That script launches the Claude Code program — the same program and the same account sign-in used throughout this project — in a “non-interactive” mode. That simply means it runs once, quietly, with no chat window, and closes itself the moment it's done.
- **3.** It hands that program one specific set of written instructions: the contents of `GMAIL_SYNC_TASK_PROMPT.md`, a plain-text file that spells out exactly what to look for in your email and exactly how it's allowed to update the tracker.
- **4.** Claude reaches your Gmail account through what's called an MCP connector — a secure, pre-approved bridge that lets it search for and open messages. It never sees or handles your password, and the connector simply doesn't offer any way to send, delete, or label email — those actions aren't available to be used, not just switched off.
- **5.** For every relevant email it finds, Claude uses the same `repository.py` rules the dashboard itself relies on to safely write the update into `applications.db` — never by editing the file directly, always through those same guarded, double-checked functions.
- **6.** Before finishing, it writes a plain-English summary of what it did into a new file in `data\sync_logs`, and updates `gmail_sync_state.json` with the time of this successful check, so the next run knows exactly where to pick up.
- **7.** The Claude Code program then exits completely. Nothing stays running in the background between checks — the computer goes back to doing nothing related to JobTrace until the next scheduled time.

You don't need to be talking to Claude for this to work

This whole chain is started directly by Windows itself, on a timer — not by you opening a chat. It reuses the sign-in already set up on this computer, the same way an email app can check for new mail on a schedule without you opening it first.

Why there's no “API key” involved

Many tools that connect to an AI do it by having a secret code — an “API key” — built into the program, which lets it call the AI directly and bills its owner for every use, automatically and often without a visible receipt. JobTrace was deliberately built to avoid that entirely: there is no API key anywhere in this project, not in the dashboard and not in the automation.

- **No key is stored anywhere in JobTrace.** Instead of an app secretly calling an AI service, the scheduled task simply opens the same Claude Code program already installed and signed in on this computer — the identical program used to build JobTrace in the first place — and gives it a task, exactly as if you'd typed a request into it yourself.
- **It rides on your existing sign-in, not a metered service.** Because it uses that regular sign-in rather than a developer API key, there's no per-request charge and no separate bill to watch — it's covered by the same Claude account and

subscription already in use, at no additional cost.

- **JobTrace's own code has no AI in it at all.** The website and the Python program behind it only ever store, calculate, and display data — they have no built-in understanding of language or email. All of the “reading and making sense of an email” intelligence happens separately, inside the Claude Code program, only during a scheduled run — never inside JobTrace itself.

A quick note on the word “API”

Earlier, this guide mentioned JobTrace's own small API — the short list of web addresses (like `/api/applications`) that the dashboard's webpage uses to talk to its own Python program on this computer. That's a completely separate, unrelated thing from an AI API: it's purely local, free, and has nothing to do with Claude. Nowhere in JobTrace — not the website, not the Python program, not the scheduled task — does anything call an external, paid AI API. The only way Claude is ever involved is as the ordinary Claude Code program, running under your own regular sign-in.